

# CSE 4345: Big Data Analytics

Week 1, Part 1 — Digital Data Landscape & Introduction to Big Data

Department of Computer Science & Engineering  
Premier University, Chittagong

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# Course & Lecture Information

## Course Details

**Code:** CSE 4345  
**Credits:** 3 (Theory)  
**Type:** Elective – Data Science  
**Pre-req:** CSE 2221 – DBMS

## Today

**Week / Part:** 1 of 11 | Part 1 of 2  
**CLO:** CLO1  
**PLO:** PLO(a)

## CLO1

*“Understand big data characteristics”*

## Part 1 Covers

- The modern data explosion
- Types of digital data
- The 3Vs and 5Vs framework
- Distributed computing basics
- Ivy League alignment
- Student assessment pool

# Lecture Agenda

- 1 The Data Revolution
- 2 Types of Digital Data
- 3 Big Data: The 3Vs and 5Vs Framework
- 4 Distributed Computing Fundamentals
- 5 Ivy League Alignment
- 6 Student Assessment Pool

# The Data Revolution

# We Are Living in the Data Age

## Data generated every 60 seconds (2024 estimates):

- **6,000** tweets posted per second on X (Twitter)
- **500 hours** of video uploaded to YouTube per minute
- **1 million** messages sent on WhatsApp
- **5 million** Google search queries executed
- **500 TB** of new data ingested by Facebook daily
- **50,000** apps downloaded from Apple App Store

## IDC Global Datasphere Forecast

By 2025, the world will generate and replicate **175 zettabytes** of data annually—a  $23\times$  increase from 2018.

## Data Volume Units

| Unit           | Bytes     |
|----------------|-----------|
| Kilobyte (KB)  | $10^3$    |
| Megabyte (MB)  | $10^6$    |
| Gigabyte (GB)  | $10^9$    |
| Terabyte (TB)  | $10^{12}$ |
| Petabyte (PB)  | $10^{15}$ |
| Exabyte (EB)   | $10^{18}$ |
| Zettabyte (ZB) | $10^{21}$ |
| Yottabyte (YB) | $10^{24}$ |

# The Fundamental Problem: Single Machine vs. Reality

## What a Single Server Can Handle (2024)

| Resource                    | Maximum   |
|-----------------------------|-----------|
| RAM (high-end server)       | ~64 TB    |
| Disk throughput (SSD)       | ~7 GB/s   |
| RDBMS comfortable limit     | ~10–50 TB |
| Time to read 1 PB at 7 GB/s | ≈41 hours |

## The Gap

Modern organisations generate data at rates that **fundamentally outpace** what any single machine can store, process, or query in acceptable time.

## The Big Data Engineering Response

- **Distribute** data across hundreds of commodity nodes
- **Move computation to data** rather than fetching data to a central processor
- **Replicate** blocks for fault tolerance (3× in HDFS)
- **Parallelise** every operation—divide, compute, merge

### Core principle (GFS, 2003):

*“Commodity hardware will fail. Design for failure from the start.”*

## Types of Digital Data

# Three Categories of Digital Data

**Structured**  
RDBMS, CSV, Excel

~20%  
of world data

**Semi-Structured**  
JSON, XML, Logs

~10%  
of world data

**Unstructured**  
Images, Video,  
Free Text

~70%  
of world data

## Key Insight

Over **80%** of the world's data is unstructured—this is the frontier that Big Data tools are built to unlock. Traditional SQL databases cannot process it without significant pre-processing.



# Structured Data

## Definition

Data organised into a **predefined schema** (rows and columns). Schema is enforced *at write time* (**schema-on-write**).

## Characteristics:

- Fixed, known schema before ingestion
- Stored in relational databases (MySQL, Oracle, PostgreSQL)
- Queryable directly with standard SQL
- Strong ACID guarantees
- Easy to index, filter, aggregate

## Sample: Bank Transaction Table

| TxID | Date       | Amt (BDT) | Type |
|------|------------|-----------|------|
| T001 | 2024-01-01 | 5,000     | CR   |
| T002 | 2024-01-01 | 1,200     | DR   |
| T003 | 2024-01-02 | 8,750     | CR   |

## Big Data relevance:

Structured data in legacy RDBMS is often the *source* that must be *migrated* into Hadoop or a data lake via **Apache Sqoop** (covered in Week 8).

## Domain Examples

# Semi-Structured Data

## Definition

Data with a **flexible, self-describing schema** embedded within the data itself. No separate schema definition is required.

## Characteristics:

- Schema carried by the data (tags, keys, attributes)
- Human-readable and machine-parseable
- Schema can evolve without breaking existing records
- Dominant format for web APIs and event logs

## Sample: Twitter API JSON Response

```
{
  "user_id": "bd456789",
  "tweet": "Big data is transforming
    Bangladesh fintech!",
  "timestamp": "2024-01-15T10:32Z",
  "likes": 142,
  "retweets": 37,
  "location": "Chittagong, BD",
  "media": null
}
```

**Big Data context:** Apache Hive and Spark can query JSON natively using **schema-on-read**—no schema pre-definition required at ingest time (covered in Week 7).

## Domain Examples

REST API responses (JSON), configuration files

# Unstructured Data—The Big Data Frontier

## Definition

Data with **no predefined schema or relational format**. Cannot be stored in row/column tables without significant transformation.

### Major categories:

- **Text:** Social media posts, news articles, clinical notes, legal contracts
- **Images:** X-rays, satellite photos, product images, CCTV frames
- **Video:** YouTube, dashcam footage, surgical recordings
- **Audio:** Call-centre recordings, podcasts, ECG waveforms
- **Sensor streams:** GPS pings, IoT telemetry, EEG

## Integrated Healthcare Record (Bangladesh Context)

A single outpatient visit generates all three types:

- **Structured:** Blood pressure, pulse, lab numeric results (RDBMS table)
- **Semi-structured:** HL7/FHIR XML discharge summary
- **Unstructured:** Physician's handwritten notes (scanned PDF), ultrasound DICOM image, voice prescription recording

All three types must be **integrated** to build a complete clinical picture—a core CLO2 challenge.

# Data Types: Comparison Summary

| Dimension      | Structured         | Semi-Structured            | Unstructured         |
|----------------|--------------------|----------------------------|----------------------|
| Schema         | Predefined (rigid) | Self-describing (flexible) | None                 |
| Storage        | RDBMS              | Files, NoSQL, APIs         | Object stores, HDFS  |
| Query tool     | SQL                | XPath, JSONPath, Hive      | NLP, CV, Spark MLlib |
| Schema timing  | On-write           | On-read                    | N/A                  |
| World share    | ~20%               | ~10%                       | ~70%                 |
| Big Data tool  | Sqoop → Hive       | Kafka, Hive                | Spark + ML models    |
| <b>Example</b> | Bank transaction   | Tweet JSON                 | MRI scan image       |

## Exam Tip

**JSON is semi-structured, not unstructured**—a common mistake. JSON carries keys (a schema) embedded within the data; an MRI image carries no embedded query schema at all.

## Big Data: The 3Vs and 5Vs Framework

# Defining Big Data—Multiple Perspectives

## Gartner / META Group Definition

*"Big data is high-volume, high-velocity and/or high-variety information assets that demand cost-effective, innovative forms of information processing that enable enhanced insight, decision making, and process automation."*

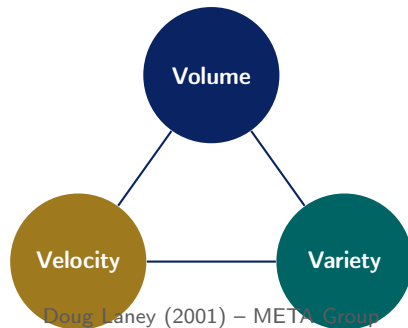
—Beyer & Laney, Gartner (2012)

## McKinsey Global Institute (2011)

*"Big data refers to datasets whose size is beyond the ability of typical database software tools to capture, store, manage and analyze."*

## Operational Definition

Big Data = data where the 3Vs exceed the capacity



The 3Vs remain the **canonical definition** of Big Data, taught in every course from Stanford CS246 to CMU 15-721.

# V1—Volume: The Scale of Data

## Definition

The **sheer quantity** of data generated, stored, and processed. Volume is the most intuitive “big” in Big Data.

## The Engineering Challenge

How do you store and query 500 TB when the largest single-instance database comfortably handles  $\approx 50$  TB?

## Production-scale examples:

- **Facebook / Meta:** 500 TB new data ingested per day
- **Google:** >100 PB processed per day (internal, 2023)
- **Walmart:** >2.5 PB customer transaction data
- **CERN LHC:**  $\approx 15$  PB of collision data per year
- **Bangladesh Election Commission:** 110 million NID biometric records
- **bKash:** Millions of micro-transactions per day

## Solution Architecture Preview

- **HDFS** (Week 3): Distribute data as 128 MB blocks across a cluster of thousands of commodity nodes
- **Replication factor 3:** Every block stored on 3 different nodes
- **Horizontal scaling:** Add nodes as data grows linearly—no downtime
- **Cloud object storage:** AWS S3, Google GCS for virtually unlimited scale

# V2—Velocity: The Speed of Data

## Definition

The **speed** at which data is generated, streamed, and must be processed or acted upon.

### High-velocity real-world examples:

- **X (Twitter):** 6,000 tweets posted per second
- **Visa payment network:** 24,000 transactions/second (peak)
- **IoT sensor networks:** 1 billion device readings per hour
- **NYSE / DSE:** Millions of stock quotes per second
- **Pathao (BD):** GPS location ping every 3 s per active driver
- **Mobile banking (bKash):** Real-time fraud scoring in  $<200$  ms

## Processing Spectrum

| Mode        | Latency  | Tool            |
|-------------|----------|-----------------|
| Batch       | Hours    | Hadoop MR       |
| Micro-batch | Seconds  | Spark Streaming |
| Stream      | Millisec | Apache Flink    |
| Real-time   | Sub-ms   | Kafka + in-mem  |

## Solution Preview

- **Apache Kafka** (Week 8): Distributed log for high-throughput event ingestion
- **Spark Streaming** (Week 6): Stateful micro-batch stream processing
- **Apache Flink:** True continuous stream



# V3—Variety: The Diversity of Data

## Definition

The **diversity** of data types, formats, sources, and update cadences that must be jointly ingested, integrated, and analysed.

## Sources of variety:

- Multiple **formats**: CSV, JSON, XML, Avro, Parquet, DICOM, MP4
- Multiple **sources**: RDBMS, REST APIs, sensors, social media, ERP
- Multiple **languages** and character encodings (UTF-8, UTF-16, Bengali)
- Multiple **update frequencies**: historical snapshots + live streams
- Multiple **semantic meanings**: a “date” field could be

## The Engineering Challenge

How do you JOIN a customer record from MySQL with their social media posts, call-centre audio transcription, and purchase-history CSV, all in one query?

## Bangladesh Government Analytics Platform

Integrating data from 50 district hospitals requires:

- MySQL RDBMS tables (structured)
- Pharmacy sales CSV files (structured)
- Community health worker XML reports (semi-structured)

# Extended Framework: The 5Vs of Big Data

| V               | Definition                                    | Real-World Example                                                |
|-----------------|-----------------------------------------------|-------------------------------------------------------------------|
| <b>Volume</b>   | Scale of data stored and generated            | Facebook: 500 TB/day ingested                                     |
| <b>Velocity</b> | Speed of generation and required processing   | Visa: 24,000 transactions per second                              |
| <b>Variety</b>  | Diversity of formats, types and sources       | CSV + JSON + DICOM + audio files                                  |
| <b>Veracity</b> | Trustworthiness and accuracy of data          | GPS signal drift, OCR errors, bot tweets, missing sensor readings |
| <b>Value</b>    | Business/societal utility extracted from data | Revenue gain, lives saved, fraud prevented                        |

## Exam-Critical Distinction

The **original 3Vs** (Lanev, 2001) addressed *systems* challenges. The **extended 5Vs** (Gartner, 2012)

# Distributed Computing Fundamentals

# Scaling Strategies: Vertical vs. Horizontal

## Vertical Scaling (Scale Up)

Add more power to **one machine**: faster CPU, more RAM, larger disks.

- Simple—no distributed-systems complexity
- Hard physical ceiling (max RAM  $\approx$  64 TB today)
- Single point of failure
- **Exponentially expensive**: a  $2\times$  more powerful server often costs  $8\times$  more

## Traditional RDBMS Approach

Oracle Exadata (vertical) can cost \$500,000+/year in licences alone.

## Horizontal Scaling (Scale Out)

Add more **commodity machines** to the cluster.

- Each node is cheap (\$3,000–\$8,000 commodity server)
- Near-linear scalability:  $2\times$  nodes  $\approx 2\times$  throughput
- Fault tolerant: losing one node does not halt the cluster
- Google's GFS cluster (2003): **thousands** of commodity PCs, each expected to fail

## Yahoo! Production Evidence (2008)

10,000 commodity nodes, 5 PB storage—cheaper than one equivalent mainframe, and far more

# The CAP Theorem (Brewer, 2000)

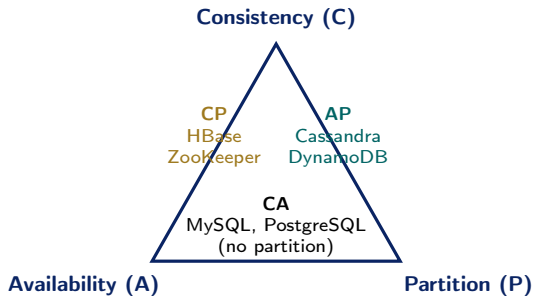
Theorem Statement (Brewer, PODC 2000; formally proved: Gilbert & Lynch, 2002)

A distributed data system can simultaneously guarantee **at most two** of:

- **Consistency**: every read returns the most recent write
- **Availability**: every request receives a response (not guaranteed to be latest)
- **Partition tolerance**: system operates despite network partitions

## Practical Implication

In any real distributed system, network partitions *will* occur. P must be tolerated. Therefore, the design



## Big Data Choices

**HBase (CP)**: used in Hadoop—strong consistency, sacrifices availability on partition.

**Cassandra (AP)**: used by Netflix—always available, eventual consistency.

# Batch vs. Stream Processing

| Dimension        | Batch Processing                          | Stream Processing                                  |
|------------------|-------------------------------------------|----------------------------------------------------|
| Data state       | Stored / historical                       | In-motion / live                                   |
| Latency          | Minutes to hours                          | Milliseconds to seconds                            |
| Throughput       | Very high (optimised for)                 | Moderate                                           |
| Fault recovery   | Re-run the batch                          | Exactly-once semantics                             |
| Typical use case | Monthly billing, ML training, nightly ETL | Fraud detection, real-time alerts, live dashboards |
| Technologies     | Hadoop MapReduce, Hive, Spark Batch       | Apache Kafka, Spark Streaming, Apache Flink        |
| Complexity       | Lower                                     | Higher                                             |

## Week 8 Preview: Architectural Choice

The **Lambda architecture** runs both layers in parallel (batch for accuracy + stream for latency). The

## Ivy League Alignment

## Course Philosophy (Leskovec, Rajaraman & Ullman)

*"The challenge of big data is not just storage—it is running algorithms that are fast enough to be useful on datasets that do not fit in the memory of a single machine. This forces a rethinking of everything."*

— Mining of Massive Datasets, Preface

## Key Benchmark: Sub-Quadratic Requirement

- On  $n = 10^9$  records, an  $O(n^2)$  algorithm requires  $10^{18}$  operations
- At  $10^9$  ops/second, that is **31.7 years** of compute time
- Big Data forces every algorithm to target  $O(n)$  or  $O(n \log n)$

## Benchmarks Across Ivy League

- **Stanford CS246**: MapReduce, Spark, LSH, graph mining, recommender systems
- **Harvard CS265**: Storage systems, query optimisation, indexing at scale
- **MIT 6.830**: Distributed consistency, transaction models
- **CMU 15-721**: Modern columnar and in-memory databases
- **Columbia COMS 4995**: End-to-end big data engineering pipelines

## CSE 4345 Positioning

This course covers the **foundational**



## Student Assessment Pool

## Instructions

Select the single best answer. Answers are **bolded** for instructor reference; present without bold to students.

- Q1. Who first formally proposed the **3Vs** framework for Big Data?
- ☐ (a) Jim Gray    (b) **Doug Laney**    (c) Jeffrey Dean    (d) Erik Brynjolfsson
- Q2. Which of the following is an example of **unstructured data**?
- ☐ (a) SQL relational table    (b) CSV file    (c) **MRI scan image**    (d) JSON API response

## Instructor Notes

**Q1:** Students commonly confuse Laney with MapReduce authors (Dean & Ghemawat). **Q2:** JSON (option d) is a common wrong answer—it is *semi*-structured because it carries key names; an MRI has no schema at all.

- Q3. The CAP theorem states that a distributed system cannot simultaneously guarantee more than two of:
- a) Cost, Availability, Performance
  - b) **Consistency, Availability, Partition Tolerance**
  - c) Correctness, Atomicity, Persistence
  - d) Concurrency, Accuracy, Partitioning
- Q4. Which “V” of Big Data specifically refers to the **speed** at which data is generated and must be processed?
- a) Volume    (b) Variety    (c) **Velocity**    (d) Veracity

### Instructor Notes

**Q3:** All four options are designed to sound plausible—requires precise recall. **Q4:** “Veracity” (option d) is the most common wrong answer since it starts with “Ve-”; students must know the precise definitions of each V.

## Instructions

State True or False. Award full marks only if the student provides a **one-sentence justification**.

| Statement                                                                                   | Answer |
|---------------------------------------------------------------------------------------------|--------|
| 1. Structured data accounts for approximately 80% of all digital data in the world.         | False  |
| 2. The 3Vs Big Data framework was extended to 5Vs by adding Veracity and Value.             | True   |
| 3. Horizontal scaling means adding more RAM and faster CPUs to a single existing server.    | False  |
| 4. Batch processing is suitable for detecting a stolen credit card being used in real time. | False  |

## Instructions

Answer each question in **100–150 words**. Use specific terminology covered in class.

- Q1. Explain the **3Vs** (and the extended 5Vs) of Big Data, providing one concrete real-world example for *each* dimension.
- Q2. Differentiate between **structured, semi-structured, and unstructured data**. Provide examples exclusively from the **healthcare domain in Bangladesh**.
- Q3. State and explain the **CAP theorem**. Why is it a fundamental constraint for designing distributed Big Data systems? Give one concrete example of a real system that makes each of the three trade-offs (CA, CP, AP).

## Instructions

Answer in **200–300 words**. Show step-by-step reasoning. Partial credit is awarded for correct reasoning even with incomplete conclusions.

### A1. 5Vs Analysis—Pathao (Bangladesh ride-sharing):

Pathao collects a GPS ping every 3 seconds from 500,000 active drivers, integrates real-time traffic feeds, surge pricing algorithms, and customer written reviews.

- Analyse this scenario through **all 5Vs**.
- Which V poses the greatest **engineering challenge**? Justify.
- What happens to the overall system if Veracity is compromised (GPS signal drifts by 200 m)?

### A2. Processing Paradigm Trade-off:

Compare batch and stream processing for: **(a)** generating a monthly bank statement, **(b)** detecting that a stolen bKash mobile account is being drained 10 seconds after theft. Justify why each scenario demands a different processing model and identify the key engineering trade-offs.

# Textbook & Reading References

## Primary Textbooks for Week 1

- **[SA]** Acharya & Chellappan. *Big Data and Analytics*, Wiley, 2015. **Chapter 1**
- **[TE]** Erl, Khattak & Buhler. *Big Data Fundamentals*, Prentice Hall, 2016. **Chapters 1, 3**
- **[MMS]** Mayer-Schönberger & Cukier. *Big Data: A Revolution*, Houghton Mifflin, 2013. **Chapter 1**
- **[LRU]** Leskovec, Rajaraman & Ullman. *Mining of Massive Datasets*, 3rd ed., Cambridge, 2020. **Chapter 1**  
*Free PDF: [mmds.org](http://mmds.org)*

## Supplementary—Covered in Week 1 Part 2 (Seminal Papers)

- Laney, D. (2001). *3D Data Management: Controlling Data Volume, Velocity and Variety*. META Group Research Note. **[PRIMARY]**
- Brewer, E. (2000). *Towards Robust Distributed Systems*. PODC Keynote.
- Gilbert & Lynch (2002). *Brewer's Conjecture and the Feasibility of Consistent Available Partition-Tolerant Web Services*. SIGACT.

# Questions & Discussion

Week 1, Part 1 | CSE 4345 Big Data Analytics

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*“Without data you’re just another person with an opinion.”*

—W. Edwards Deming

Next Lecture: Week 1, Part 2

Seminal Research: Laney (2001) · CAP Theorem Formal Proof  
Case Study: McKinsey Global Institute (2011) · Live Data Exercise